IRELE

Mobile and wireless networks

Research paper analysis

Authors:

Arico Amaury

Mototani Yuta

Tchakounte Loic

Sipakam Cedric

Professor:

Dricot Jean-Michel

TA:

Ladner Navid

Academic year:

2025 - 2026

Contents

List of Figures

1

Introduction

2

System model

3

Proposed protocols

4

Attack model and protocol resilience

The section focuses on the resilience of the key generation protocol discussed during the previous chapters. Our attacker, Eve, is assumed to be a passive eavesdropper for both communication architechtures. For each scenario, we will discuss the level of information Eve can extract from her observations and derive the upper bound on the probability to have a successful eavesdropper attack. While mathematical developments in this paragraph rely on information theory tools such as conditional entropy and mutual information, our objective is not to reproduce the paper detailed mathematical derivations but rather provides a intuitive interpretation of those. In particular, we will emphasize how the information leakage securities (quantization, channel decorrelation and the privacy amplification) take parts in the probability upper bound calculation.

4.1 Mutual information

We first need to look into the signal seen by Eve to grasp the potential danger. The equation 4.2 expands mathematically the signal as combination of element-wise multiplication.

$$w_{i,\text{ed},1} = s_i \circ v_i \circ h_{i,\text{ae}} + n_{i,\text{ae}}, \tag{4.1}$$

$$w_{i,\text{ed},2} = s_i \circ v_i \circ h_{i,\text{be}} + n_{i,\text{be}}. \tag{4.2}$$

One can understand that the observation being successful will depend entirely on the correlation between the channel $h_{i,\text{ae}}$ and $h_{i,\text{ab}}$ or between the channel $h_{i,\text{be}}$ and $h_{i,\text{ab}}$.
We discussed previously about the spatial correlation of the channel and the reciprocity properties between the channel observed by the sender and the receiver. It would be interesting to quantify the correlation between the channel seen by our attacker Eve and Bob-Alice|Alice-Bob channel as it intervenes in the probability computation of leaked information seen from Eve side.
From channel communication course, we know that the magnitude of a channel observed at distance d, can be built as a statistical model realisation (from Rice or Rayleigh pdf model). The spatial correlation between

two channels realisation considering small scale fading can be expressed as a Bessel function. Mathematically it boils down to the equation 4.3.

$$J_0(kd) \approx \begin{cases} 1, & d \ll \lambda, \\ \\ 0, & d \approx 0.38\,\lambda, \\ \\ \text{oscillatory decay}, & d > \lambda. \end{cases} \tag{4.3}$$

Intuitively, this result shows that the wireless channels observed by two receivers separated by more than approximately half a wavelength (general rule of thumb) are weakly correlated. As a consequence, an eavesdropper located at such a distance experiences an essentially independent channel realization. This spatial decorrelation, confirming the channel randomness seen by a sufficient distant observer, is the fundamental property exploited for physical layer key generation.

We also know that channel coefficients are complex numbers. From the statistical models, channel can be therefore decomposed in an imaginary sub-channel and real sub-channel independently and identically distributed as $\mathcal{N}(0,\sigma^2)$. From 4.4, the real and imaginary parts of Bob|Alice and Eve are correlated with ρ.

$$\rho = [J_0(kd)]^2 \tag{4.4}$$

From the channel correlation and the channel model distrubition, we can build the covariance matrices of both channels, h_{ae} and h_{ab} and the covariance matrice from joint channel $(h_{\text{ae}}, h_{\text{ab}})$, the former being used to calculate the mutual information in equation 4.5.

$$\mathbf{C}_{\text{Bob}} = \begin{pmatrix} \frac{\sigma_{\text{b}}^2}{2} & 0 \\ 0 & \frac{\sigma_{\text{b}}^2}{2} \end{pmatrix} \tag{4.5}$$

$$\mathbf{C}_{\text{Eve}} = \begin{pmatrix} \frac{\sigma_{\text{e}}^2}{2} & 0 \\ 0 & \frac{\sigma_{\text{e}}^2}{2} \end{pmatrix} \tag{4.6}$$

$$\mathbf{C}_{\text{Joint}} = \begin{pmatrix} \frac{\sigma_{\text{b}}^2}{2} & 0 & \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} & 0 \\ 0 & \frac{\sigma_{\text{b}}^2}{2} & 0 & \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} \\ \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} & 0 & \frac{\sigma_{\text{e}}^2}{2} & 0 \\ 0 & \frac{\rho\sigma_{\text{b}}\sigma_{\text{e}}}{2} & 0 & \frac{\sigma_{\text{e}}^2}{2} \end{pmatrix} \tag{4.7}$$

The mutual information, denoted $I(h_{\text{ae}}, h_{\text{ab}})$, can be expressed through the conditional entropy development in equation 4.8

$$I(h_{\text{ae}}, h_{i,\text{ab}}) = H(h_{\text{ae}}) + H(h_{\text{ab}}) - H(h_{\text{ab}}, h_{\text{ae}}) \tag{4.8}$$

Where the entropies exhibit the level of randomness or uncertainty inherent to the channels.
By expressing the entropies as multivariate gaussian distrubition, the mutual information results into the expression 4.12.

$$\begin{aligned}
I(h_{\text{ae}}, h_{i,\text{ab}}) &= H(h_{\text{ae}}) + H(h_{\text{ab}}) - H(h_{\text{ab}}, h_{\text{ae}}), && (4.9)\\
&= \frac{1}{2}\log\big(\det(2\pi e\mathbf{C}_1)\big) + \frac{1}{2}\log\big(\det(2\pi e\mathbf{C}_2)\big) - \frac{1}{2}\log\big(\det(2\pi e\mathbf{C}_3)\big), && (4.10)\\
&= \log\big(\pi e\sigma_{\text{b}}^2\big) + \log\big(\pi e\sigma_{\text{e}}^2\big) - \log\big((\pi e\sigma_{\text{b}}\sigma_{\text{e}})^2(1-\rho^2)\big), && (4.11)\\
&= -\log(1-\rho^2) && (4.12)
\end{aligned}$$

The final equality convey that the level of information leakage does depend exclusively of the spatial correlation between the attacker channel and the communication channel, and therefore independent of the absolute channel power or, in other words, independent of the signal to noise ratio.

4.2 Upper bound probability of successfull attacks

The mutual information defined, we have all the tools to compute the upper bound probability of successfull attacks. For that, we must compute each increase of probability generated by the implemented security processes, namely the quantization process, the channel spatial correlation and the privacy amplification process. In a first time, we will consider the direct communication architecture and afterwards compute the upper bound for the communication through untrustred relay.

4.2.1 Direct communication

the probability of a successful Eve's eavesdropping attack is upper bounded as stated by the equation 4.13.

$$Pr(success) < (2^{-2\delta} + \sqrt{2I(h_b, h_e)})^N + 2^{-N\delta}. \tag{4.13}$$

With :

1. $2^{-2\delta}$ term providing the probability of successfull guess of the binary representation after quantization. The factor 2 coming from the imaginary and real channel coefficient decomposition.
2. $\sqrt{2I(h_b, h_e)}$ increasing the probability of recovering the right bits given Eve's observation from her own channel.
3. the exponent N being the OFDM subcarriers number, each carrier seeing a narrowband flat channel.

4. $2^{-N\delta}$ term raising the upper bound due to the additional guess of the key generation during the privacy amplification process.

It is worth mentioning that the privacy amplification increases the upper bound probability of successfull attack but removes the information leaked during the code-secure sketch process.

4.2.2 Communication through untrusted relay

As a remainder, in this communication architecture, the relay, Carol, is sending back the received signal amplified by a public factor α. The attacker, Eve, has access to **two** signals shwon in the equations 4.14 and 4.15.

$$w_{i,\mathrm{e},1} = s_i \circ h_{i,\mathrm{ae}} + v_i \circ g_{i,\mathrm{be}} + n_{i,\mathrm{e1}}, \tag{4.14}$$

$$e_{i,3} = \alpha\big(s_i \circ h_i + v_i \circ g_i + n_{i,2,\mathrm{r}}\big) \circ f_{i,\mathrm{ce}} + n_{i,\mathrm{e2}}. \tag{4.15}$$

From the OFDM preambles symbols, the channel $f_{i,\mathrm{ce}}$ can be estimated leading to the recovery of the signal $w_{i,\mathrm{e},2}$ in 4.16.

$$w_{i,\mathrm{e},2} = s_i \circ h_i + v_i \circ g_i + n_{i,2,\mathrm{r}} \tag{4.16}$$

The spatial correlation property of the channel is, in this case, not providing the same security against the eavesdropper attack as the direct communication. The reason is Eve can virtually be considered as having access to the same level of information as the relay. This comes from the fact that the signal $w_{i,\mathrm{e},2}$ can be retrieved completely. The mutual information contains in $I(w_{\mathrm{i},ab}; w_{i,\mathrm{e},1}, w_{i,\mathrm{e},2})$ is much more significant than in the *section 4.2.1*.As such, the upper bound cannot be express from the equation 4.13 as saturating to 1 and thus not giving relevant information about the level of security for the relay architecture. Therefore, the idea is to pass by the Fano'sequality developed in the equation 4.18.

$$\Pr(C_N) \leq \big(1 - \frac{H(q_a \mid w_{i,\mathrm{e},1}, w_{i,\mathrm{e},2}) - 1}{\log_2 |\mathcal{Q}_A|}\big)^N, \tag{4.17}$$

$$\Pr(C_N) \leq \big(1 - \frac{H(q_a) - I(w_{\mathrm{i},ab}; w_{i,\mathrm{e},1}, w_{i,\mathrm{e},2}) - 1}{\log_2 |\mathcal{Q}_A|}\big)^N. \tag{4.18}$$

If the attacker, Eve, cannot deduce the key during the communication session, the remaining probability to rightly guess the hashed key is equal to $2^{-N\delta}$.The total upper bound probability is therefore stated in the equation 4.19.

$$\Pr(C_N) \le \Big(1 - \frac{H(q_a) - I(w_{\mathrm{i},ab}; w_{i,\mathrm{e},1}, w_{i,\mathrm{e},2}) - 1}{\log_2 |\mathcal{Q}_A|}\Big)^N + 2^{-N\delta}. \tag{4.19}$$

5

Results and discussion

The results section focuses on the bit generation rate (BGR) and the bit mismatch rate (BMR), as these metrics enable a direct comparison between the proposed algorithm and existing secret key generation protocols but also state the protocol limitations in real conditions. We will conclude by a critical analysis of the protocol and its limitations.

The research paper compares the results obtained by three different scenarii. For all the sceanrii, the channel used OFDM parallel transmission with 16 subcarriers and QAM symbols, the quantization after transmission is fixed at $\delta = 2$. The first scenario is a direct reciprocal channel communication. The second is a non light of side channel with realistic 5G mmWave channels coefficients (from the NYUSIM network simulator) in urban micro-cellular environment operating in 28 GHz with very low outdoor-indoor loss and Eve distant of 10 meters from Bob and Alice. The last one is relay-based architecture with only direct communication with the relay.

5.1 Bits generation rate

For all the scenarii, in the current modulation, quantization configuration and bandwidth division, the bit generation rate adds up to 64 bits/OFDM symbol. The paper mentions **64 bits/packet** expressing that the transmitted QAM symbols are average over all the OFDM symbols ensuring a minimization of the noise, optimizing the BMR per communication session. The key is generated after 4 sessions, current transmitted bits being added modulo 2 with the previous one to enhance secrecy (relevant for NLOS inside communication - scenario 2). Regarding the performance, the paper protocol achieves comparable level of bits generation rate as protocols developed for dynamic environment. For static environment, previous protocols could generate between 0.5 bit per packet to 8 bits per packet.

For completeness, we suppose we want to generate a 128 bits secret key for the current 5G wireless network. For communication operating at 28GHz, we find 14 OFDM symbols per packet. Assuming a subcarrier spacing of 120 KHz, a 128 bits secret key can be established in few milliseconds, as shown in the equation 5.5. Thus, the key generation timing is equivalent to computation of key encryption methods such as RSA or handshaking methods and well within the timing constraint for 5G New Radio communication.

$$T_{\text{sym}} = \frac{1}{\Delta f} = \frac{1}{120\,\text{kHz}} \approx 8.33\ \mu\text{s}, \tag{5.1}$$

$$T_{\text{sym}}^{(\text{CP})} \approx 9\ \mu\text{s} \qquad\qquad \textbf{with cycle prefix}, \tag{5.2}$$

$$T_{\text{slot}} = 14 \times T_{\text{sym}}^{(\text{CP})} \approx 14 \times 9\ \mu\text{s} = 126\ \mu\text{s}, \tag{5.3}$$

$$T_{\text{32-bits}} = 4 \times T_{\text{slot}} \approx 4 \times 126\ \mu\text{s} \approx 0.5\ \text{ms}, \tag{5.4}$$

$$T_{\text{128-bits}} = 4 \times T_{\text{32-bit}} \approx 2\ \text{ms}. \tag{5.5}$$

5.2 Bits mismatch rate

The figure 5.1 illustrates the results for bits mismatch rate between Alice and Eve and compares it with the BMR between Alice and Bob. The comparison spans all the three setups. As remainder, the bits mismatch rate is the ratio between the computed quantized bits after transmission and the corresponding true quantized bits. We can observe that the BMR between Alice and Bob decreases with the SNR, while the BMR between Alice and Eve exhibits a very slow reduction behaviour and converges to a *plateau*. Nevertheless, the second scenario BMR between Alice and Eve is significantly lower than the other setups, decreasing in the SNR range from 5 to 15 dB before also reaching a *plateau*.

This moderate BMR in the indoor NLOS communication is attributed to sparse, or clustered scattering of the signal. The spatial correlation model of the physical channel rests on the assumption of infinite amount of multipath components (MPC) or scatters resulting to Rayleigh or Rice fading statistics. In 5GNR communication, the scatters are characterized by limited diffusion and dominant reflections, resulting to a small number of dominant paths. Under these sparse scattering conditions, the probability that Eve observes the same cluster of scatters as Alice and Bob is significantly higher than in the idealized case of infinite MPCs environment, and thus explaining the low BMR for this scenario.

Nevertheless, even for the NLOS communication, the BER between Alice and Eve converges to 50% after privacy amplification, thereby confirming the security robustness of the proposed protocol - Figure 5.2.

5.3 Limitations

The weakness of the proposed protocol relies on key assumptions resulting to potential limitations in case of protocol deployement.

1. **Passive eavesdropper attack :** The protocol assumes a passive eavesdropper attacker, not interfering with the communication between Alice and Bob. In case of active attacks from Eve, such as jam-

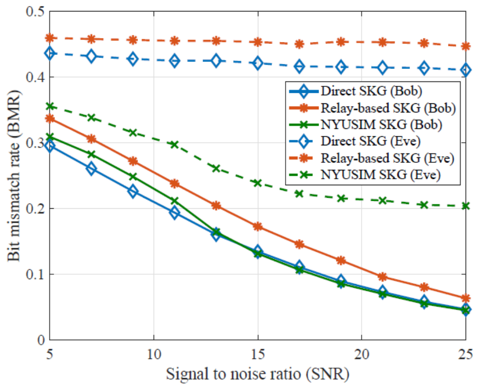

Figure 5.1: Bits mismatch rate - all setups comparison.

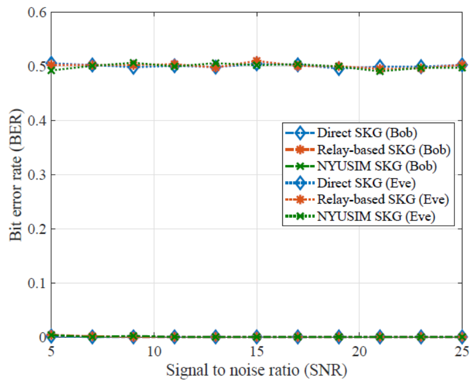

Figure 5.2: Bits mismatch rate - all setups comparison.

ming, the correlation is degraded, preventing Alice and Bob from reliable recovery of the introduced randomness from both parties.

2. **Channel stability :** The key generation process has been proved to be robust in static environment and remains valid for slow varying channels as long as the time coherence of the channel exceeds the key generation time. However, if the time needed to process the secret key is longer than the channel stability duration, the observations of Alice and Bob decorrelate, resulting to an increase of the BMR and potential key generation failure.

3. **Hardware synchronisation :** The protocol assumes a good synchronisation of Alice's and Bob's hardwares. Imperfect syncrhonisation may require concession on quantization resolution or on the rate of error-correcting code should be made ensuring successfull correlation and key derivation.

6 AI discussion

The usage of the AI in the framework of this report focuses on three pillars.

1. **Intuition behind the mathematical developments :** The major usage of the AI was centered around an assistant role for decrypting unknown mathematical simplification or decomposition. The mathematical fondations regarding the channel physics or information theory are well known for the electronics engineers. Nevertheless, some details like the Fano's inequality or the Bessel function of first order was just slightly brushed in our course without a deep interrogation about the meaning. The AI in this case could give us detailed undertanding and lead us to the intuition hidden behind the mathematics.

2. **Help about the report construction :** The second idea was to use the AI as support tool for the direction to give to this report. More specifically, it was to confirm the result or analysis-oriented report with a focus on the intuition behind the mathematical expressions instead of just summarize the research paper.

3. **Support for example construction and validation of results :** Finally, AI was used as a tool to construct simplified numerical examples providing support in the discussion of the scenarii results or for the critical analysis. For instance, the example about the key generation time was supported by the artificial intelligence to confirm the number of OFDM symbols contained in a 5GNR communication packet and double-checked in the litterature afterwards.

7 Conclusion